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ORIGINAL ARTICLE

Parameter optimization for laser welding of dissimilar aluminum alloy: 5052-H32 and 6061-T6 considering wobbling technique

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Abstract
The wobbling technique which consists of oscillating the laser beam at high frequency and in a specific pattern is an innovative way to enhance the quality of weld. The wobbling widens the area covered by the laser beam resulting in a wider seam that provides improved mechanical properties. The study investigated the effect of parameters such as the laser power, the gap between the two butt-welded plates, and the amplitude of oscillation using Taguchi's method by setting a L9 design of experiments. The laser power level has a low range as the spot diameter is kept at a very small size to increase the power density and improve the energy efficiency of the laser welding process. The results revealed the optimal welding set-up and conditions to enhance the mechanical properties of welds. The value of laser power and the amplitude is optimized relatively to the ultimate tensile strength and the weld shape expressed by the depth to width ratio (D/W) that is correlated to the ductility. A fitness function combined with the steepest descent method applied to regression analysis allows figuring out the optimal parameters defined by the value of laser power at 1118 W and an amplitude of 0.8 mm resulting in a predicted welded part with 206.8 MPa of tensile strength and 4.5% of ductility.

Keywords Beam wobbling · Laser welding · Aluminum alloys · Taguchi and ANOVA method · Microstructure characterization · Mechanical property

Nomenclature

AA	Aluminum alloy
Adj MS	Adjusted mean squares
Adj SS	Adjusted sum of squares
Al	Aluminum
ANOVA	Analysis of variance
ASTM	American Society for Testing and Materials
BM	Base metal
D/W	Depth to width ratio
DoF	Degree of freedom
EDS	Energy dispersive X-ray spectroscopy
FZ	Fusion zone
h	Defect measurements
HAZ	Heat-affected zone
HBF4	Tetrafluoroboric acid
HV	Vickers hardness
K	Degree Kelvin
kW	Kilowatt
L	Liter
m	Meter
Mg	Magnesium
min	Minute
mm	Millimeter
MPa	Megapascal
N_{amp}	Amplitude normalized value
N_{LP}	Laser power normalized value
N_{uts}	Ultimate tensile strength normalized value
OM	Optical microscope
R-sq	Coefficient of determination
s	Second
SCFH	Standard cubic feet per hour
Seq SS	Sequential sum of squares
Si	Silicium
SiC	Silicon carbide
t	Thickness
UTS	Ultimate Tensile Strength
V	Volt
W	Watt
w_1	Importance weight of tensile strength

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w_2	Importance weight of laser power
w_3	Importance weight of amplitude
$\propto$n	Iteration interval steps
µm	Micrometer

1 Introduction

Among the lightest metals used in manufacturing, aluminum has attracted a growing interest in several sector of industries such as shipbuilding, ground transportation, and aerospace mainly because of its lightweight advantage that leads to the reduction in the vehicle weight and low energy consumption [1, 2]. AA5052-H32 is highly solicited in the automotive industry for the manufacturing of miscellaneous parts because of its good machinability and high strength [3]. AA 5052-H32 is also known for being a corrosion-resistant aluminum alloy, a characteristic this alloy shares with AA6061-T6 alloy, but the weldability of the latter is poor even though its ultimate strength is higher than of AA5052-H32 [4]. It is the reason both are often assembled as a dissimilar welding to take the advantages offered by each alloy and obtain a stronger part [5]. Nonetheless, after being welded, aluminum tends to lose about 20 to 40% of its mechanical strength [6]. For that reason, many research works have emerged to develop techniques for the enhancement of the utilization of aluminum in manufacturing [7]. Aluminum presents difficulties to be welded mainly because of its properties. The high thermal conductivity of the aluminum remains a real issue for its weldability because a powerful heat source is required to generate enough energy so the aluminum alloy can be welded [8]. It is problematic for welding techniques such as arc welding because high heat input may result to a high thermal distortion [9]. Among the popular welding technique, laser welding is one of the best alternatives for aluminum alloy. Indeed, the laser welding comprises several benefits such as narrow fusion zone, small heat-affected zone, and less heat distortion as well as accessible noncontact process that make it one of most effective choice [10]. Furthermore, the small spot size used in the technique allows to concentrate the energy in a small area, so the power density is high enough to generate heat that can weld aluminum alloy [11]. In terms of productivity, the laser welding can be executed at high speed, and the process can be optimized and automated involving artificial intelligence [12]. Nonetheless, even for the laser beam process, several challenges have been reported during aluminum welding such as the solidification crack due to the large solidification range, solidification shrinkage, thermal contraction, and lack of liquid metal flowing in the interdentritic space during phase change as well as the formation of porosities due to gas entrapment or keyhole instability [13, 14]. An innovation has increased the interest in laser welding, which is the wobbling technique which consists of oscillating the laser beam at high frequency during welding [15]. Barbieri et al. [16] conducted a study on this technique and showed a weld quality improvement within the oscillation mode for laser welding of AA6082-T6 to AA5754-H111 in lap joint. They concluded that the use of oscillation from wobbling increases the weld width at the metal interface resulting into an increase of the tensile strength. Wobbling limits the heat generated since the oscillation of the beam distributes the energy during the welding process. This phenomenon limits the inconvenience of laser welding such as alloying elements loss or thermal distortion. Ramiarison et al. [17] investigated the improvement of the weldability of aluminum alloys thanks to the combination of high-power density laser beam and the wobbling technique. They concluded that the wobbling limited the heat-generated defects and enhanced the ductility of the AA 5052-H32. Wang et al. [18] studied the effects of different beam oscillating modes on 5A06 aluminum alloy sheets and demonstrated that the novel infinity beam oscillation presented less welding defects and better mechanical properties compared to the linear and circle oscillating modes [18]. A similar study draws the same conclusion in which circular and sinusoidal shapes were tested to determine better to reduce the thermal gradient and obtain an equiaxial microstructure for stronger mechanical performance [19]. All those works illustrate that the welding of materials has reached a high level within the emergence of innovative techniques such as wobbling. However, much remains to be explored since simply oscillating the beam is not enough to achieve optimal welding improvements. In that respect, it is necessary to study the effect of parameters and get more comprehension of phenomenon during the laser welding using the wobbling technique especially for welding of dissimilar alloys [20]. A such study has been made by Zhang et al. [21] by the investigation of welding speed effect on the hot cracking for dissimilar welding of AA5754 and AA6013 in which they found that a quicker velocity tends to increase the susceptibility to hot cracking. Unfortunately, the study of Zhang et al. is limited at the welding speed. For wobbling technique, apart from the oscillation mode, parameters such as the frequency and the amplitude of oscillating have been studied by researchers. Di Matteo et al. [22] investigated the effect of amplitude on dissimilar laser welding of aluminum and copper and found that high amplitude increases the weld width and the ultimate tensile strength. Li et al. [23] concluded that the high frequency combined with large amplitude reduces the amount of porosity in AA 5083 and the oscillation enhances the mechanical property of weld. Wang et al. [24] found the same conclusion with 6061-T6.

Even though those research works constitute an important advance in understanding the wobbling technique, the parameters considered only concern oscillation parameter, and none of them has an application in aluminum dissimilar alloy welding. To our knowledge, only Zhang et al. [21] included parameters from laser welding process; however, their study is only focused on welding speed and oscillation diameter. Chen et al. [25] also carried out a thorough analysis of both welding parameter and oscillation parameter effects on the dissimilar welding of AA 6061-T6 and 2A12-T4. Their study discussed the influence of those parameters on the weld shape and the weld segregation, but the discussion was not pushed in deeper to consider the mechanical properties of the weld.

Regarding to that state of the art, in the present study, laser beam wobbling technique was used for autogenous butt joint laser welding of dissimilar AA 5052-H32 and AA6061-T6 aluminum sheets. The objectives are to investigate the effect of parameters on the weld quality and determine the optimal values of parameters to achieve the best quality of AA5052-H32 and AA 6061-T6 dissimilar welding. The laser welding parameters, i.e., laser power, beam wobbling amplitude, and the gap between two coupons, were set according to Taguchi L9 orthogonal array for the analysis. The choice of those parameters is justified by the lack of study, especially for the laser power which is not invoked among the cited research. As outputs from the series of experiments, the welds' geometrical and mechanical properties were taken as responses, and ANOVA method was performed to evaluate the effect of parameters. A regression model was developed, and a gradient descent algorithm was applied to extract the optimum parameter combination for the best weld quality.

2 Material and method

Two different sheets of aluminum alloys AA6061-T6 and AA5052-H32 (chemical compositions shown in Table 1) were used as the base metal for a butt joint autogenous laser welding. The dimensions of each individual Al sheets were 300 mm (length) × 100 mm (width) × 2 mm (thick). The laser welding was performed using a IPG YLS 6000 laser source with a wavelength of 1070 nm and small spot diameter of 0.1 mm to compensate for the low range of laser power by increasing the power density, exploiting the results of Kancharla et al. [26], stating that the efficiency of the wobbling technique is amplified when a small spot diameter is used. The device is connected to the D50 IPG wobbling laser welding head and mounted on a Kuka robot. A circular pattern shown in Fig. 1 is chosen following the recommendation of Wang et.al. [24] and Hagenlocher et al. [19], stating that the circular pattern has given, respectively, a sounder weld and more desirable microstructure. During the whole welding experiences, pure argon was used as shielding gas with constant flow rate of 50 SCFH (23.60 L/min). Laser welding experiments were designed by the Taguchi method using a L-9 orthogonal array as shown in Table 2. Samples dedicated to microstructure analysis are cut, mounted, and grinded using SiC papers up to 1200 grit, polished using 3 and 1 µm diamond suspensions, and then etched in Barker's electrolytic etch (5%HBF4 solution) at 30 V for 2 min to

Table 1 Composition of aluminum alloys

Elements	Cr	Cu	Fe	Mg	Mn	Si	Zn	Ti	Al
5052-H32	0.15–0.35	0.1	0.4	2.2–2.8	0.1	0.25	0.1	-	Rem
6061-T6	0.04–0.35	0.15–0.4	0.7	0.8–1.2	0.15	0.4–0.8	0.25	0.15	Rem

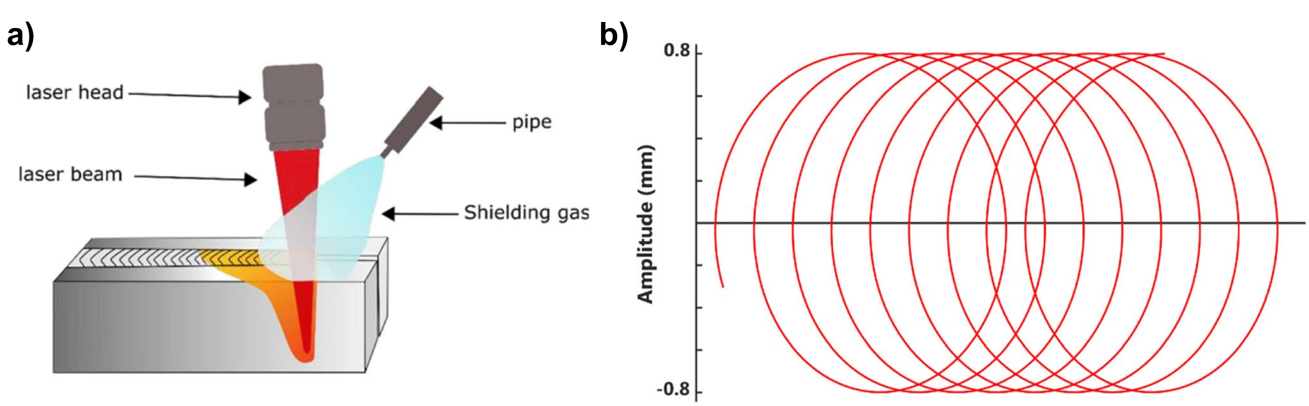


Fig. 1 **a** Experimental set-up illustration. **b** Circular wobbling path

Table 2 Taguchi L9 orthogonal array

Test	Laser power (W)	Amplitude (mm/s)	Gap (mm)
1	1100	0.6	0
2	1100	0.7	0.1
3	1100	0.8	0.2
4	1150	0.7	0
5	1150	0.8	0.1
6	1150	0.6	0.2
7	1200	0.8	0
8	1200	0.6	0.1
9	1200	0.7	0.2

Table 3 Specimens measured width

Test	Width (mm)
1	1.397 ± 0.008
2	1.344 ± 0.011
3	1.504 ± 0.015
4	1.498 ± 0.002
5	1.552 ± 0.01
6	1.440 ± 0.071
7	1.609 ± 0.083
8	1.338 ± 0.006
9	1.502 ± 0.021

reveal the grain structure. The microstructure of the welds was examined using an optical microscope (OM) (Olympus BX51M). The composition of a welded part is obtained from the energy dispersive X-ray spectroscopy (EDS). Hardness is also measured along the weld width using Vickers microhardness machine Clemex CMT.HD®. Tensile tests were conducted according to ASTM E8 standard. Three samples are machined from the welded coupons to comply with the geometry, and the test is realized with Material Testing System 810.

3 Weld geometry, microstructure, and mechanical properties

3.1 Weld geometry and imperfection

The width measurement for all welding conditions is summarized in Table 3. The weld geometrical imperfections and defects identified in the weld's cross-sections were underfill, excessive penetration, undercut, and microporosity. Very fine microcracks were also optioned in one welding condition that will be explained later. Underfill imperfections are shown in Fig. 2 as well as the root reinforcement featured by an excessive penetration of the weld relatively to the material thickness. The maximum acceptable underfill depths and root undercut according to ISO 13919–2 level B and level C standard are, respectively, 0.1 mm ($h \leq 0.05$ t) and 0.2 mm ($h \leq 0.1$ t) for the both mentioned imperfection types. The results of surface underfill, root reinforcement,

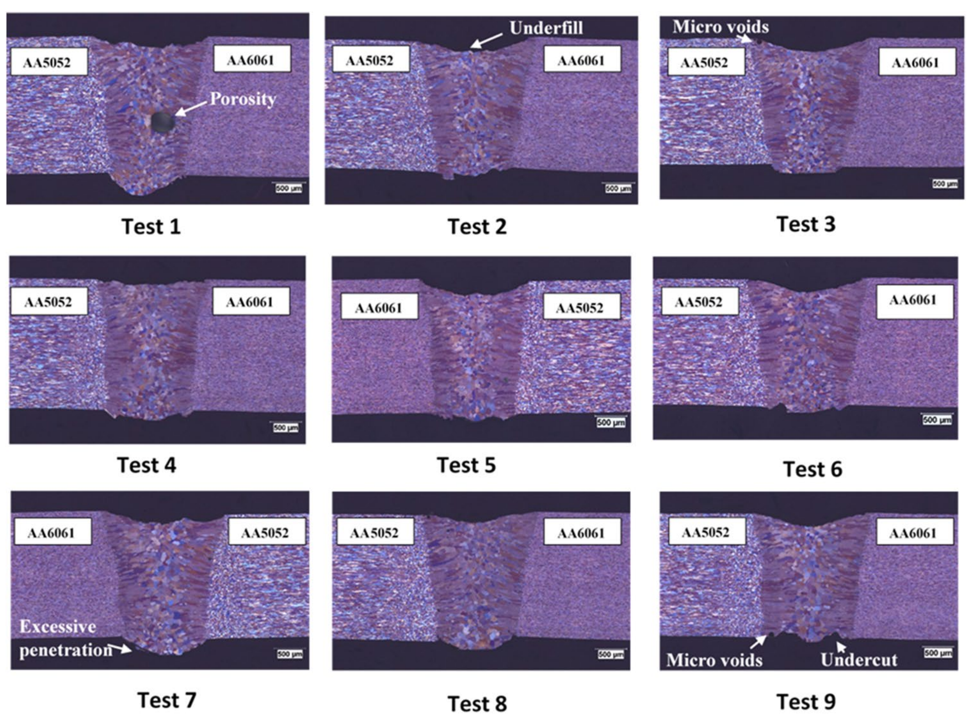


Fig. 2 Cross-section of tested specimens

Table 4 Defect quality level for each test according to ISO 13919–2 (D, moderate; C, medium; B, elevated)

Test	Measurements			Defect quality level		
	Underfill (mm)	Undercut (mm)	Root reinforcement (mm)	Underfill	Undercut	Root reinforcement
1	0.190	-	0.365	C	B	B
2	0.198	0.060	0.158	C	B	B
3	0.223	-	0.123	D	B	B
4	0.093	0.076	0.165	B	B	B
5	0.170	-	0.222	C	B	B
6	0.226	0.085	0.151	D	B	B
7	0.193	-	0.275	C	B	B
8	0.154	-	0.163	C	B	B
9	0.209	0.097	0.075	D	B	B

and undercut depths for all welding conditions are presented in Table 4. Regarding underfill depth, most of the welding conditions met level C acceptance criteria of the ISO 13919–2 standard except tests 3, 6, and 9 which met level D. Those tests refer to the welding conditions with 0.2-mm gap size, and the deepest underfill is for test 6 which refers to the maximum gap of 0.2 mm and minimum amplitude of 0.6 mm. Test 4 is the only welding condition that presents a non-significant underfill categorized as level B. In terms of undercut imperfection at root side, all the welding conditions met level B acceptance criteria of the ISO 13919–2 standard with maximum undercut depth for test 9 measured at 0.097 mm. The root excessive penetration was another discontinuity observed in the welds' cross-sections. The excessive penetration height for all welding conditions in this study met the maximum limit according to ISO 13919–2 level B (0.5 mm, given by $h \leq (0.2\ \text{mm} + 0.15t)$ as presented in Table 4. Porosity is a supplementary imperfection that was detected in weld transverse cross-sections. Test 1 showed a large porosity appearing in the center of the weld seam with a diameter of 0.35 mm. The large porosity in the path of keyhole is most likely due to unstable keyhole collapse [7]. Microvoids of less than 100 µm diameter were also observed in the weld seam close to the fusion line close to the top or bottom surfaces that probably are hydrogen-induced porosity.

Microstructure shows the presence of microcracks in fusion zone for test 3 and test 6 depicted in Fig. 3a and b at high magnifications. These microcracks are intergranular with less than 100 µm that forms a weld fusion zone center and root excessive reinforcement. These cracks are most likely shrinkage solidification cracks that form due to metallurgical factors as well as the level of local strain yielded at the terminal stage of solidification. If there is not sufficient liquid metal to heal the opening space created by the thermal/shrinkage strain during solidification, microcracks will form along the grain boundary [27]. In our case, the cracks start to appear when the gap is larger considering that no filler metal is added to compensate the lack of material in the gap. An observation drawn by Zhang et al. [21] as well as Mirakhorli et al. [28] finds a worsening of crack formation when the gap is applied in aluminum welding. The amplitude of oscillation is also smaller for test 6 which is favorable for crack formation as described by Li et al. [23], stating that a large amplitude at frequency of 200 Hz is prompted to give a weld with less porosities and cracks. The level of laser power which dictates the heat input can also influence the appearance as discussed by Ma et al. [29]

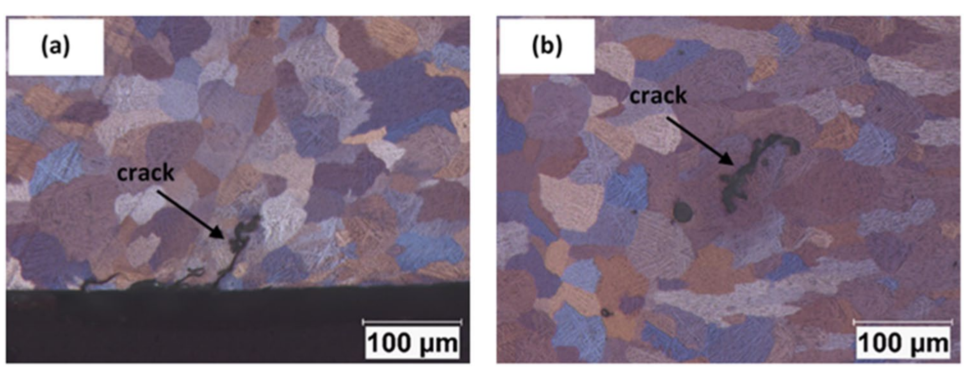


Fig. 3 Microcracks in fusion zone **a** root of test 3 and **b** center of test 6

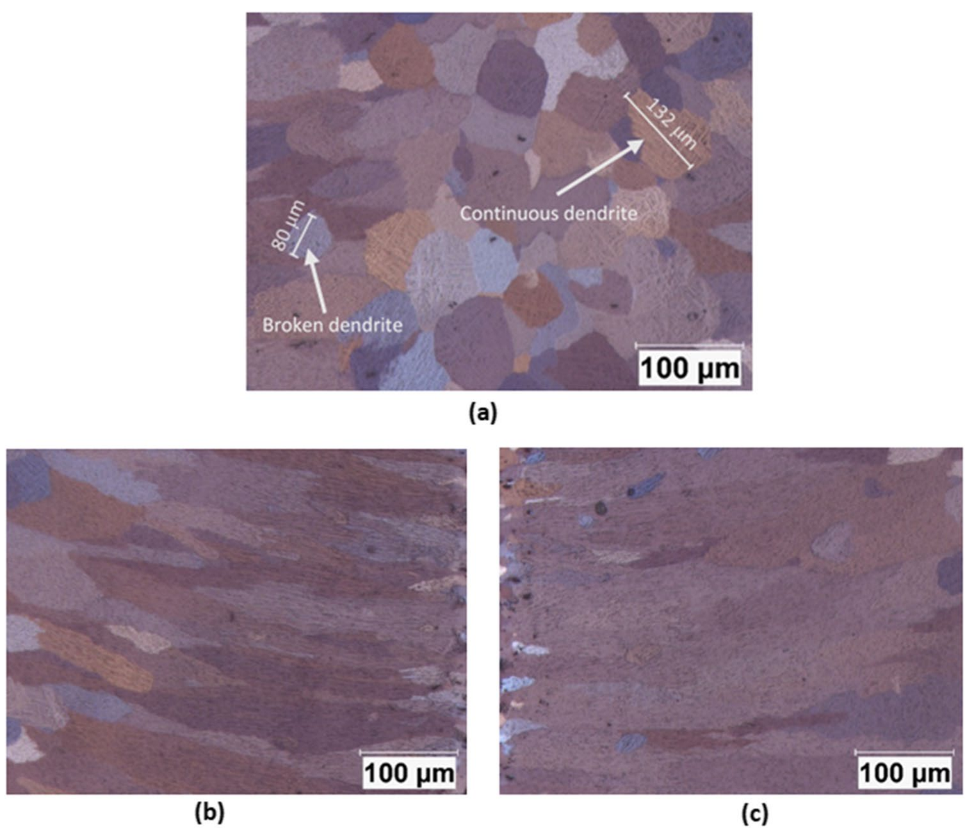
stating that heat improves the ductility of Mg favoring the crack inhibition.

3.2 Microstructural analysis

Figure 4a–c displays the microstructure in the transverse section of the FZ under an optical microscope. The fusion zone is characterized by elongated columnar grains close to fusion boundaries in both AA6061-T6 and AA5052-H32 sides of the fusion zone, as shown in Fig. 4b and c and transforms into small equiaxed grains as approaching the center of the weld (Fig. 4a). Such grain structure is common in aluminum welding; the columnar grains at the fusion line boundary are formed from the epitaxial growth as the base metal acts as a substrate for the crystal nucleation [30]. In the opposite, in the center line of the fusion zone, equiaxed grains are promoted because the thermal gradient diminishes when getting away from the fusion line boundary, giving a favorable condition for the appearance of the equiaxed grains which block at the same time the columnar growth when the sufficient size and amount are reached [19, 30]. The fusion zone illustrated in Fig. 4a contains a network of dendrites characterized by long and continuous dendrites combined with broken dendrites forming an isolated and interrupted sequence among the networks. Wang et al. [24] explain the presence of those interrupted dendrites by the capability of the wobbling technique to break the dendrite evolution causing the appearance of small dendrites among the entire network present in the fusion zone.

Surrounding the fusion zone, the heat-affected zone is characterized by the phenomenon of grain coarsening due to the thermal sequence that affects the zone without phase change. Figure 5 shows the microstructure of the heat-affected of the two distinct alloys. Figure 5a shows the heat-affected zone of the AA5052-H32 also etched with Weck's reagent and characterized by α-Al precipitates taking place at grain boundaries similarly observed by Cao et al. [31] in dissimilar welding of AA5052-H32 with press-hardened steel. Moreover, the shape of grains in the 5052-H32 denoted the loss of cold work when it is compared to grains in the base metal which are flat and elongated (Fig. 6a and 7), a typical shape of grains of cold-worked aluminum alloy [32]. The heat-affected zone of the AA6061-T6 is formed by coarse grains and dispersed large precipitation Mg_2Si visible in Fig. 5b. This results are the same as found by Fadaeifard et al. [33], explaining the phenomenon by the grain overaging due to the prolonged heat exposure of the alloy during the welding process.

Fig. 4 Microstructure observation for Test 4 (**a**) fusion zone center featured by equiaxed grain and dendrites (**b**) 6061-T6 fusion side and (**c**) 5052-H32 fusion side



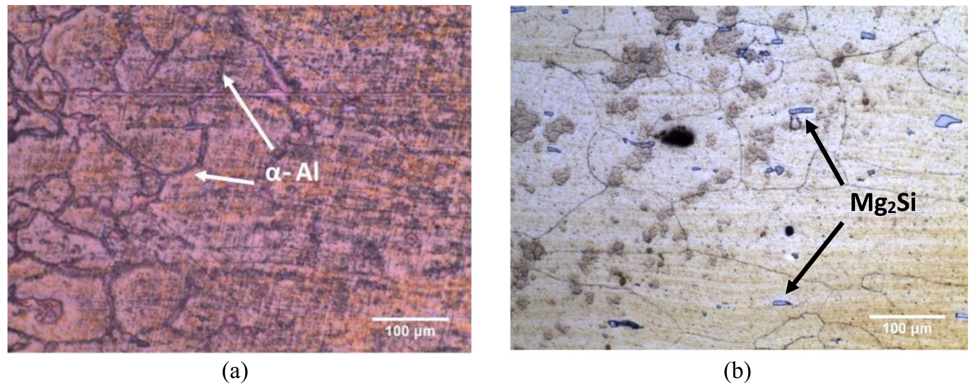


Fig. 5 Precipitates observed in the HAZ **a** AA 5052-H32 and **b** AA 6061-T6

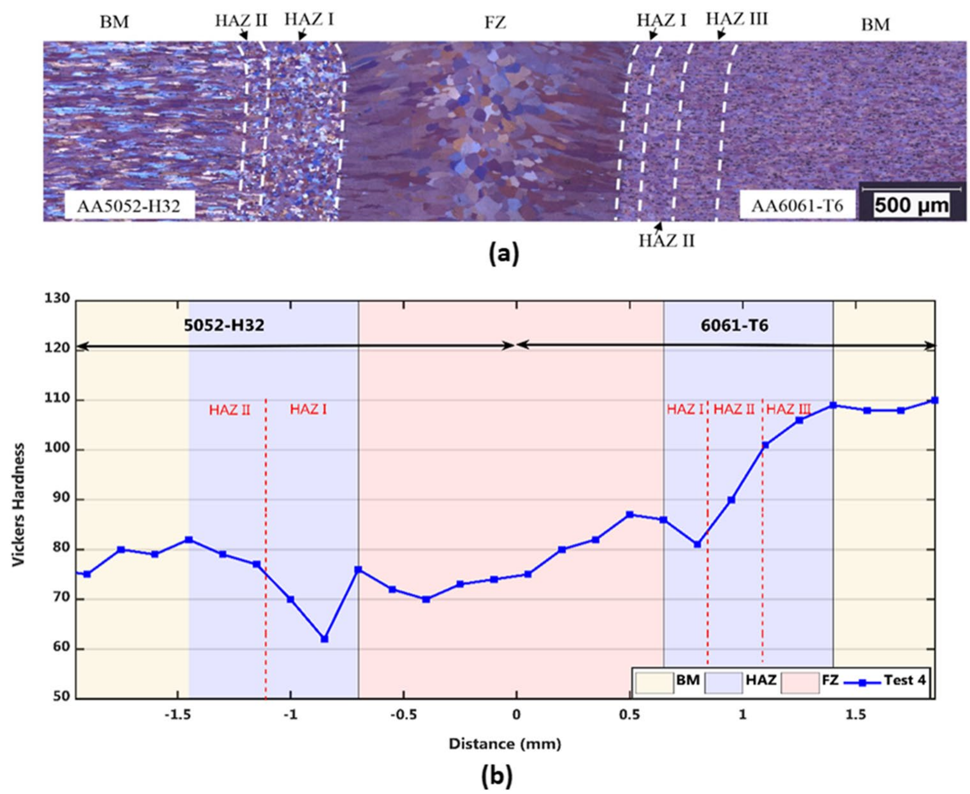


Fig. 6 **a** Miscellaneous weld thermal zones. **b** Hardness profile for dissimilar welding (test 4)

3.3 Microhardness characterization

Hardness profiles can be seen in Fig. 6b showing three zones: the fusion zone (FZ), the heat-affected zone (HAZ), and the base metal (BM). At first glance, a hardness drop can be observed regarding to the inherent hardness property of the base metal of both alloys. This result is common for aluminum welding because the thermal cycle resulting from heat input during the process conducts the metal into a change in its microstructure and composition. Generally, for the AA 6061-T6, the HAZ I is harder than HAZ II since there are more precipitates formed in the former. The HAZ III is the least affected part, and the hardness profile in Fig. 6 reinforces that observation as the heat-affected zone next to the base metal presents non-significant drop regarding to the HAZ next to the FZ. The hardness decrease can be also the results of the dilution of β” precipitates during the phase transformation leading to the appearance of GP zones in accordance to the precipitation sequence of AA 6061-T6 [19, 34]. For AA 5052-H32, the hardness drop in the HAZ can be associated to the loss of cold work as it is a non-treatable alloy, and the microstructure observation clearly shows that the grains are recrystallizing. In the fusion zone, the hardness increases, limiting the degree of hardness drop in the BM. This hardness profile is comparable to those observed by Barbieri et al. [16] who also conducted a study

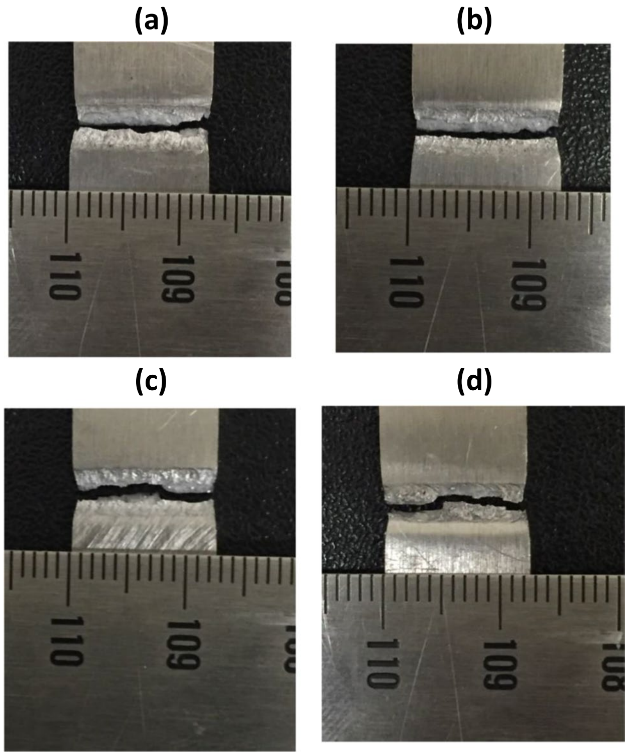


Fig. 7 Sample breakage **a** surface Test 1, **b** root Test 1, **c** surface Test 9, **d** root Test 9

on dissimilar welding with AA 5754 and AA 6082, especially in the heat-affected zone. This is the results of the overaging of the metal in this zone conducting to the coarsening of grains complying with the description of HAZ I and HAZ II in the microstructure analysis [35]. In the fusion zone, the hardness drop is mainly due to the loss of low melting point, alloying elements during the laser welding of aluminum alloys [18, 27, 36]. That is confirmed by the EDS X-ray analysis performed in FZ, HAZ, and BM for the sample in Table 5 in which the weight percentage of Mg and Si significantly decreases, respectively, 2.22% to 1.51% and 1.19% to 0.37% in fusion zone center compared to the composition of the two base metals. The loss of Mg and Si can lead to hardness reduction in FZ as those elements are known, especially Mg, to provide an important strengthening and improvement of work-hardening.

3.4 Tensile test results

Figure 8 shows strain–stress curves at the highest stress of whole tests conducted in this study. It can be observed that the curves seem to have the same shape, i.e., the elastic zone of each curve presents similar slope, suggesting that the Young modulus is not affected within the experiments. Nonetheless, in terms of UTS, there is a variation between different trials as summarized in Table 6. Test 1 shows the highest UTS of 207.5 MPa, and the weakest weld is test 9 with UTS of 195.5 MPa. A difference that can be expected as parameter settings of test 1 and test 9 is really shifted in terms of parameter level. The fracture path of tensile samples after rupture shows that the fracture mostly occurred in the fusion line probably due to the defects such as underfill, undercut, and root reinforcement that acts as stress concentration location and initiate the crack propagations. Nonetheless, there are exceptions for test 8 and test 9 in which the failure took place in the fusion line and deviated toward fusion zone as exhibited in Fig. 7c and d, complying with the fact that those are the two weakest welding condition according to tensile test results. The lowest tensile strength of test 9 can be attributed to the importance of the defects generated during the welding process revealed by the macrography analysis as the sample presents a quite high level of underfill and undercut (Fig. 2).

4 Effect of parameters

4.1 Effect of parameters on the weld width

As expected, the width of the weld is closely related to the amplitude of the beam oscillation as it increases (Table 7).

Table 5 EDS X-ray results

	Alloy	Side	wt (%)		
			Mg	Si	Al
Base metal	5052-H32	-	2.21	-	95.39
	6061-T6	-	1.38	1.19	95.34
Heat-affected zone	5052-H32	Top	2.08	0.19	95.01
	5052-H32	Root	2.28	-	94.39
	6061-T6	Top	1.45	0.42	95.25
	6061-T6	Root	1.73	0.37	95.34
Fusion zone	-	Top	1.51	0.16	95.48
	-	Root	1.52	0.37	95.85

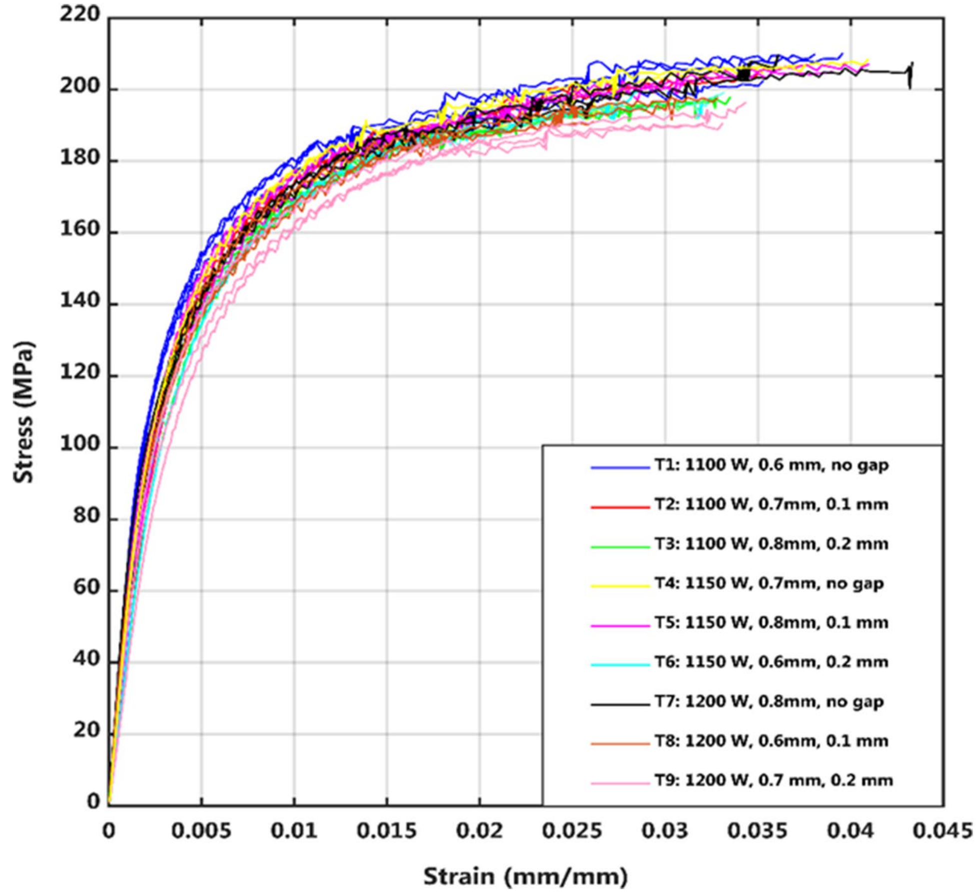


Fig. 8 Strain–stress curve for repeated experimental units' tensile test

Table 6 Taguchi orthogonal L9 design of experience with tensile test results

Test	Laser power (W)	Amplitude (mm/s)	Gap (mm)	UTS (MPa)	El. (%)
1	1100	0.6	0	207.5 ± 4.4	5.1 ± 0.8
2	1100	0.7	0.1	204.2 ± 1	3.8 ± 0.68
3	1100	0.8	0.2	197.5 ± 1.4	4.1 ± 0.17
4	1150	0.7	0	205.2 ± 4.6	4.7 ± 0.61
5	1150	0.8	0.1	204.9 ± 2.3	4.4 ± 0.42
6	1150	0.6	0.2	198.6 ± 1.3	4.3 ± 0.1
7	1200	0.8	0	205.6 ± 5.6	5.1 ± 0.71
8	1200	0.6	0.1	195.5 ± 3.8	3.7 ± 0.61
9	1200	0.7	0.2	192.6 ± 3.3	3.9 ± 0.15

Logically, the path of the laser beam is guided by the beam oscillation; therefore, as the amplitude increases, the beam path covers a larger area, widening the width of the molten zone and consequently the weld width. Research carried out on the wobbling amplitude claimed the same results [15, 16] stating that the wobbling amplitude tends to increase the weld width and Wang et al. [37] experienced that at a frequency under 200 Hz, the effect of amplitude on weld width becomes more important which is the case in this study where the frequency is kept at 150 Hz. The effect of gap is non-significant as demonstrated by the ANOVA table of the weld width. Indeed, the gap does not change the area covered by the laser beam but the amount of metal in contact with the laser beam. The main effect graph shows that

Table 7 ANOVA for weld width

Source	DoF	Seq SS	Contribution	Adj SS	Adj MS	*F* value	*P* value
Laser power	2	0.011480	16.80%	0.011480	0.005740	5.59	0.152
Gap	2	0.013468	19.72%	0.013468	0.006734	6.55	0.132
Amplitude	2	0.041300	60.47%	0.041300	0.020650	20.10	0.047
Error	2	0.002055	3.01%	0.002055	0.001027		
Total	8	0.068303	100.00%				

a higher level of laser power enlarges the width of the weld that can be explained by the buffering of the area covered by the laser beam when heat is generated alongside the wobbling path.

4.2 Effects of parameters on UTS

Table 8 shows that the most influential parameter is the gap, and it affects the tensile strength in a negative way meaning that a higher gap diminishes the tensile strength. It is related to the amount of melted metal during the welding process as described in the discussion about the effect of parameters on the weld width. As less metal is going to a liquid phase meaning that less metal constitutes the weld seam, the gap affects the weld in two ways. First, it increases the distance between plates, suggesting that there is more space to fill with phase changing metal. Second, the distance increased between plates leads to the reduction of the quantity of molten metal; hence, in addition to enlarge the space to be filled, the gap limits the amount of base material available to heal the weld. The contribution of the laser power in the mechanical property is quantified by the percentage of contribution of 23.22%. In general, the laser power dictates the heat input during welding. In the case of this study, an upper level of laser power tends to weaken the weld in terms of tensile strength as observed in Fig. 9. This occurrence can be associated to a higher heat input due to an increase of laser power. Overheating conducts to harmful consequence for weld mechanical properties such as defect susceptibility or alloying element loss as discussed by Chu et al. [30] concluding that the heat input is consequent to tensile strength and the strongest weld is obtained at low heat input and increasing the latter tends to weaken the welded parts. The effect of amplitude is minor despite the results in Fig. 9 showing that the tensile strength augments with the amplitude.

Table 8 ANOVA for UTS

Source	DoF	Seq SS	Contribution	Adj SS	Adj MS	*F* value	*P* value
Laser power	2	52.240	23.22%	52.240	26.120	2.89	0.257
Gap	2	146.216	64.98%	146.216	73.108	8.08	0.110
Amplitude	2	8.451	3.76%	8.451	4.226	0.47	0.682
Error	2	18.095	8.04%	18.095	9.048		
Total	8	225.002	100.00%				

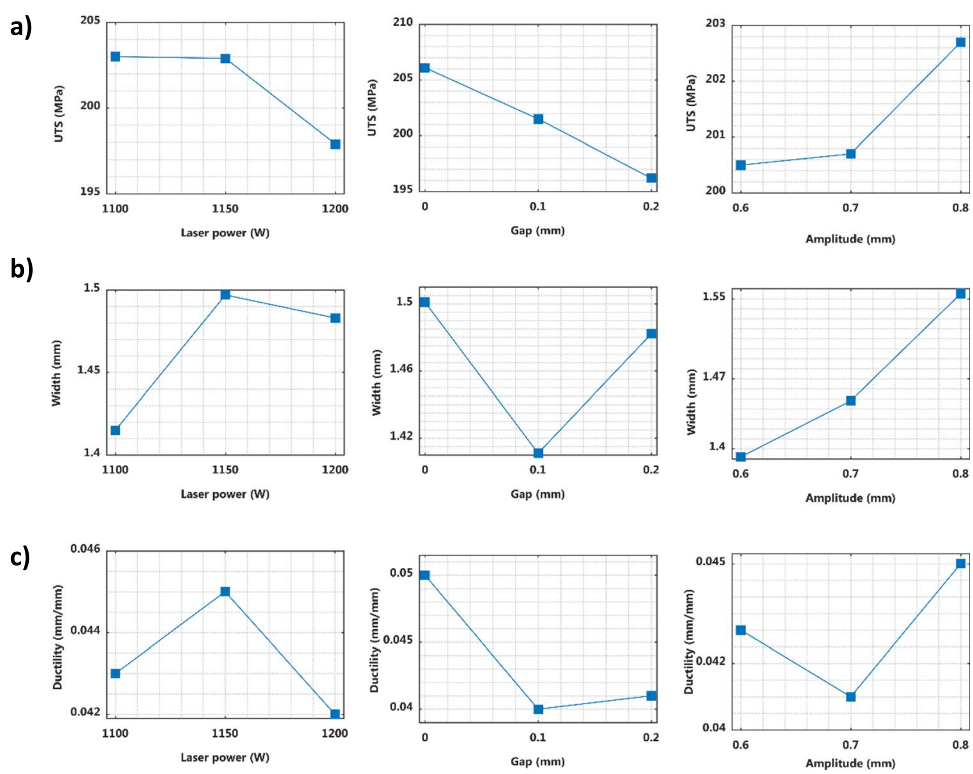


Fig. 9 Parameter effects on the outputs' **a** ultimate tensile strength, **b** weld width, and **c** ductility

4.3 Effect of parameter on the ductility

The analysis of the parameter impact on the ductility reveals that the gap significantly influences this mechanical property, and the gap augments the tendency of the weld to be more fragile. As explained previously, the presence of gap reduces the amount of the molten metal conducting in the lack of alloying elements that contribute to enhance the mechanical properties of aluminum alloy. The effect of amplitude is not significantly represented, but it is more noticeable than the laser power as shown in Table 9 in which their contributions are, respectively, 10.14% and 4.97%. For comparison, the contribution of the gap in the ductility of weld is 78.04%.

5 Numerical analysis

5.1 Output correlation

This part of the study fixes as objectives to find a correlation between those outputs. The influence of the weld shape on the weld mechanical properties has been investigated by Sinha et al. [38] who concluded a correlation between weld width and tensile strength. In the case of this study, an additional output is considered which is the depth which is for this case, the defects (underfill, root reinforcement, and undercuts) are subtracted to the thickness of the plate (Eq. 1), and the result for each test is set in Table 10. The results of the analysis are made up with the correlation of the weld shape defined by the depth to weld ratio (D/W) and the mechanical properties of the weld, namely, the tensile strength and the ductility:

$$depth = thickness\ plate - underfill - root\ reinforcement - undercut \quad (1)$$

The results show two divergent correlations in Fig. 10. The correlation between the tensile strength and the D/W is very weak (R-sq = 0.38), whilst there is a negative correlation between the ductility and the D/W with R-square of 0.73, suggesting that low weld shape ratio likely results to more ductile welded parts.

5.2 Regression fitting model

From experimental results, the predictive models are developed to forecast the tensile strength, the weld depth, and width from input parameters summarized by the following equation, respectively, for ultimate tensile strength (UTS), the weld depth, and the weld width:

$$UTS = 602 - 0.351P - 38G - 482A + 0.428PA \quad (2)$$

Table 9 ANOVA for ductility

Source	DoF	Seq SS	Contribution	Adj SS	Adj MS	*F* value	*P* value
Laser power	2	0.000011	4.97%	0.000011	0.000006	0.73	0.578
Gap	2	0.000174	78.04%	0.000174	0.000087	11.48	0.080
Amplitude	2	0.000023	10.19%	0.000023	0.000011	1.5	0.400
Error	2	0.000015	6.80%	0.000015	0.000008		
Total	8	0.000223	100.00%				

Table 10 Depth of welded specimens

Test	1	2	3	4	5	6	7	8	9
Depth (mm)	1.475	1.614	1.684	1.696	1.638	1.568	1.562	1.713	1.746

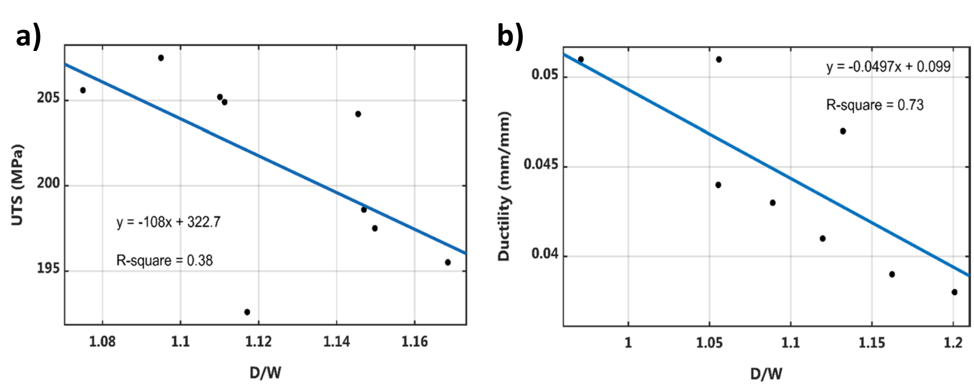


Fig. 10 Outputs' correlation. **a** D/W vs UTS. **b** D/W vs ductility

Table 11 Regression model for UTS

Source	DoF	Seq SS	Adj SS	Adj MS	F value	P value
Regression	4	204.381	204.38	51.095	9.91	0.024
Error	4	20.621	20.621	5.155		
Total	8	225.002				
	R-sq = 90.8%		R-sq (adj) = 81.7%	R-sq (pred) = 52.4%		

Table 12 Regression model for depth

Source	DoF	Seq SS	Adj SS	Adj MS	F value	P value
Regression	4	0.045388	0.045388	0.011347	3.03	0.154
Error	4	0.01496	0.01496	0.003742		
Total	8	0.060355				
	R-sq = 75.20%		R-sq (adj) = 50.40%	R-sq (pred) = 3.8%		

Table 13 Regression model for width

Source	DoF	Seq SS	Adj SS	Adj MS	F value	P value
Regression	3	0.053677	0.053677	0.017892	6.12	0.040
Error	5	0.014626	0.014626	0.002925		
Total	8	0.068303				
	R-sq = 78.59%		R-sq (adj) = 65.74%	R-sq (pred) = 25.01%		

$$Depth = -14.12 + 13.57\times10^{-3}P + 21.14A - 0.013G - 18.2\times10^{-3}PA \quad (3)$$

$$Width = -6.71 - 5.06\times10^{-3}P - 8.61A + 8.2\times10^{-3}PG \quad (4)$$

The analysis of regression demonstrated a real consistency of the model developed on the tensile strength as observed in Table 11 where the P value is < 0.05 precisely 0.024. For the depth and the width, the P value is, respectively, 0.154 and 0.04 (Tables 12 and 13). Figure 11 shows experimental results and predicted value for each output; the R-square calculated to evaluate the fitness of each model reveals high value varying from to 0.752 to 0.9089, meaning that the error rate is not significant suggesting a good reliability of the predictive model; and even the predicted R-square possess values that prevent the model from overfitting present good reliability. No model has been developed for the ductility even it comprises among the outputs from experimental results. This absence is due to the poor fitness of the model obtained from the regression, but the ductility is still considered in the predictive model as a close correlation to the depth to width ratio exists. Therefore, in the continuity of this study, the ductility will be predicted by the linear correlation with D/W expressed by the ratio from Eqs. (3) and (4).

5.3 Response surface method

The parameters plotted for the response surfaces are the laser power and the amplitude of the wobbling to give the predicted ultimate tensile strength and the D/W obtained from the developed models. For the two response surfaces, no-gap welding is considered following the study carried out in the previous part showing that keeping the gap at zero provides a wider and stronger weld. Figure 12 pointed a region where at a certain value of laser power and amplitude, the highest tensile strength can be achieved. The optimal result stands at the low laser power in the range of 1100 W and 1120 W combined with small amplitude about 0.65 mm. Nonetheless, the second response does not match with the first because the low D/W ratio can only be achieved with high amplitude combined with high laser power, or inversely, by the combination of low heat input with small amplitude. Knowing that less power can provide a better weld is encouraging because the power of laser is a significant factor in the cost-effectiveness of the laser process. However, for the amplitude, it is preferable to keep it wide weld because it has a positive influence in the mechanical properties of the weld as seen in the main effect graph displayed in Fig. 9.

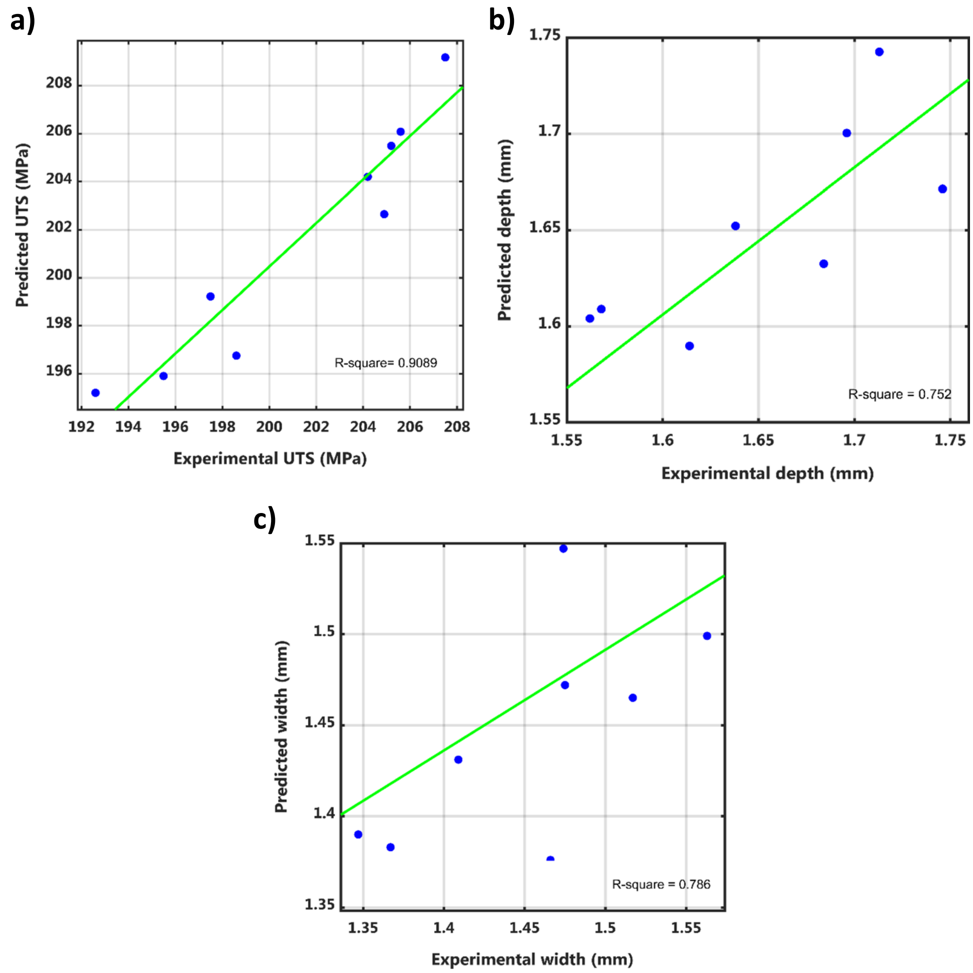


Fig. 11 Experimental results vs predicted outputs. **a** UTS. **b** Weld depth. **c** Weld width

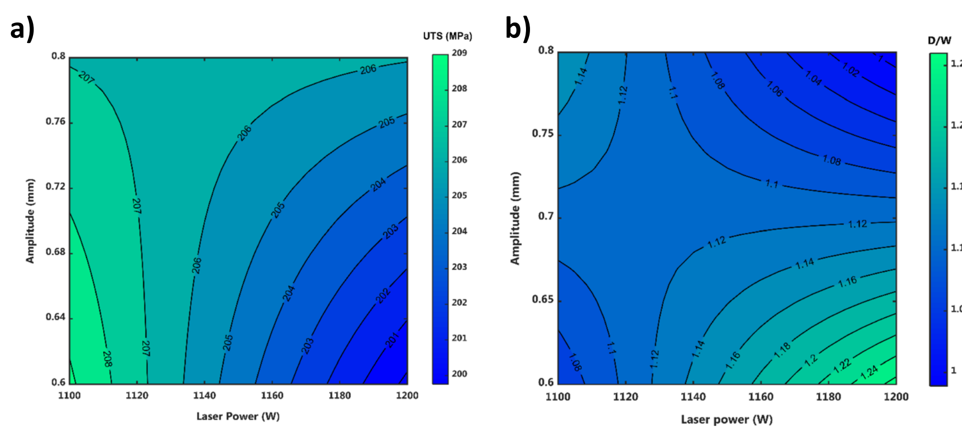


Fig. 12 Response surface for **a** the prediction of ultimate tensile strength no-gap welding set-up and **b** prediction of depth to width ratio no-gap welding set-up

6 Parameter optimization

6.1 Fitness function and constraint

The optimization consists of finding the values of parameters that give the best outputs related to the goals to be reached and the constraints that imposed by the laser welding process. In this study, the outputs are essentially made up of mechanical properties and weld shape. Obviously, the main goal is to achieve the best mechanical property that affects the effectiveness of the weld. Therefore, improving the ultimate tensile strength is considered as the most important objective of the optimization. In addition to the tensile strength, the optimization also considers the ductility of the weld. As seen in the previous part of the study, the ductility is correlated to weld shape expressed by the ratio of the weld depth by the weld width (D/W). It was observed that the lower is the ratio, the more ductile is the weld; therefore, over certain value of D/W, the weld has a larger elongation before the breakage. Regarding to the correlation study, a constraint from the predicted geometry of weld is retained, meaning that the depth to width ratio must be inferior to the value of 1.1 (D/W < 1.1). The optimization consists of maximizing the value of the fitness function expressed by the following equation:

$$Fitness\ value = w_1 N_{UTS} + w_2 N_{LP} + w_3 N_{Amp} \tag{5}$$

where N_{UTS}, N_{LP}, and N_{Amp} are, respectively, the normalized value of ultimate tensile strength, laser power, and amplitude from the predictive model equation in the range of maximum and minimum obtained from experience. The normal values (Fig. 13) are given by the following equations for each term considered in the fitness function:

$$N_{UTS} = \begin{cases} 1.2 & if\ UTS \geq 209 \\ 1 & if\ 207.5 < UTS < 209 \\ \frac{UTS-192.7}{207.5-192.7} & if\ 192.7 < UTS < 207.5 \\ 0 & if\ UTS \leq 192.7 \end{cases} \tag{6}$$

$$N_{LP} = \begin{cases} 1 & if\ LP \leq 1100 \\ \frac{1200-LP}{1200-1100} & if\ 1100 < LP < 1200 \\ 0 & if\ LP \geq 1200 \end{cases} \tag{7}$$

$$N_{Amp} = \begin{cases} 1 & if\ Amp \geq 0.8 \\ \frac{Ampl-0.6}{0.8-0.6} & if\ 0.6 < Amp < 0.8 \\ 0 & if\ Amp \leq 0.6 \end{cases} \tag{8}$$

w_1, w_2, andw_3 are, respectively, the weight of N_{UTS}, N_{LP}, and N_{Amp} which express the level of importance of those fitness function parameters. As mentioned, the ultimate tensile strength is the most important output to optimize by reaching the maximum value; that is the reason it is weighed at 80 in the fitness function. The laser power and the amplitude are arbitrarily weighed at the same importance of 10. Even if the laser power could be considered more valuable but the fact that the range of the laser power is not very large (1100 W to 1200 W), the effect on the cost-effectiveness of process would not be consequent, whereas the amplitude enlargement provides numerous advantages as described previously. The power of laser is reached to be as low as possible because using less power meaning less energy; therefore, it will be advantageous for the energy efficiency of the laser welding process. In the opposite, the amplitude is subject to maximization because of its positive influence on the weld width that tends to strengthen the weld [16].

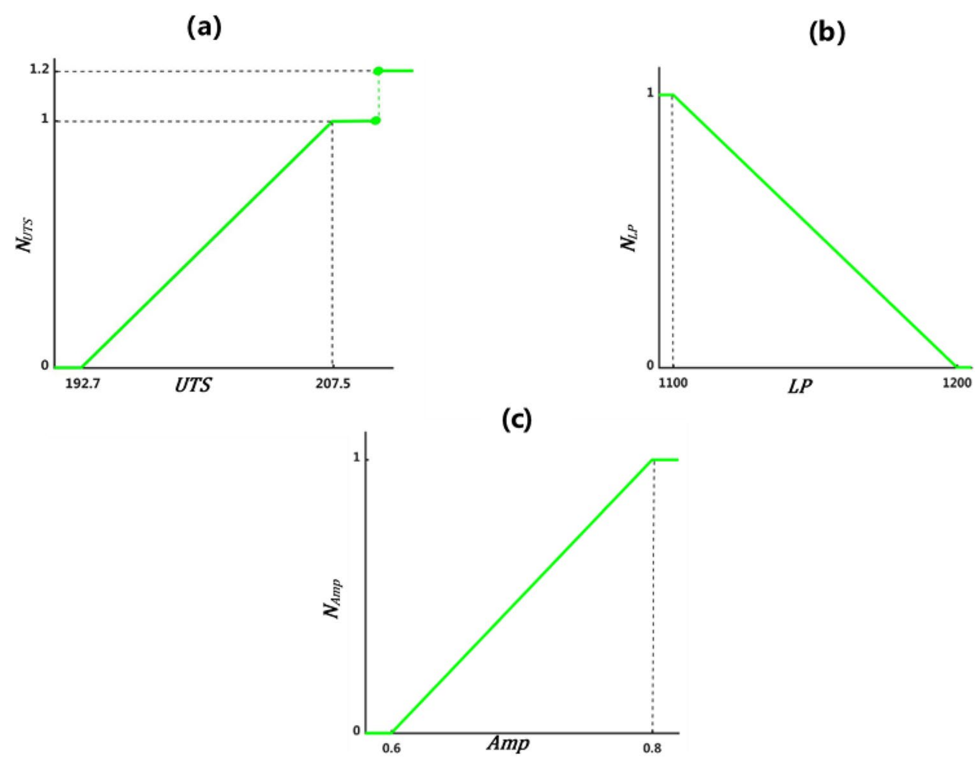


Fig. 13 Parameter normalization for the optimization fitness function. **a** Ultimate tensile strength. **b** Laser power. **c** Wobbling amplitude

6.2 Optimization algorithm

The algorithm used in the optimization is the steepest gradient method defined by the following iterative equation:

$$x_{n+1} = x_n - \alpha_n \nabla f(x_n) \tag{9}$$

$$\alpha_n = argminf(x_n - \alpha \nabla f(x_n)) \text{ with } \alpha \geq 0 \tag{10}$$

where $f(x)$ is the computed function, namely, the fitness function giving the fitness, value defined by Eq. (5) and $\nabla f(x)$ is the gradient of the fitness function. However, as mentioned above, just achieving the highest fitness value is not enough as constraint is considered (D/W < 1.1). At first, the algorithm is applied to find the optimal point for the highest fitness value. Then, a batch of points that are close to optimal point are selected, and their depth to width ratio is calculated using the predictive model for the weld shape defined by Eqs. (3) and (4). A second batch is obtained by retaining only those with D/W inferior to 1.1 in which the optimum point(s) is found. In Fig. 14, all the values with a D/W inferior to 1.1 are plotted with the fitness values obtained from the iterative process following the defined algorithm. It can be noticed that not all high fitness values have a weld shape complying with depth to width ratio conducting to a more ductile weld.

Those welded with lower laser power are likely to provide a high fitness and respect the threshold value of the depth to width ratio. That can be explained by the fact lowering the power of laser tends to increase the fitness as excepted from Eqs. (5) and (7).

Table 14 Optimized parameters

No	Laser power (W)	Amplitude (mm)	Fitness value	D/W
1	1118	0.80	94.39	1.088
2	1117	0.80	94.36	1.088
3	1119	0.80	94.31	1.089
4	1117	0.79	94.31	1.09
5	1116	0.78	94.29	1.089
6	1116	0.77	93.88	1.093
7	1117	0.76	93.35	1.096

6.3 Optimization results

Table 14 summarizes the final group of points respecting the optimization goal and constraint. Seven possible combinations are presented in the group with very close fitness values. The highest fitness value reached is 94.39 by setting the laser power at 1118 W with an amplitude of 0.8 giving 1.088 of depth to width ratio. This combination will achieve a tensile strength of about 206.8 MPa, and by applying the correlation between D/W and ductility, the weld will have an elongation break of 4.5%. Comparatively to the results obtained from the other welds using an amplitude of 0.8 mm in the design of experiments, namely, tests 3, 5, and 7, the mechanical strength from the optimized parameters is better knowing that those cited welding conditions gave, respectively, an ultimate tensile strength of 197.5 MPa, 204.9 MPa, and 205.6 MPa.

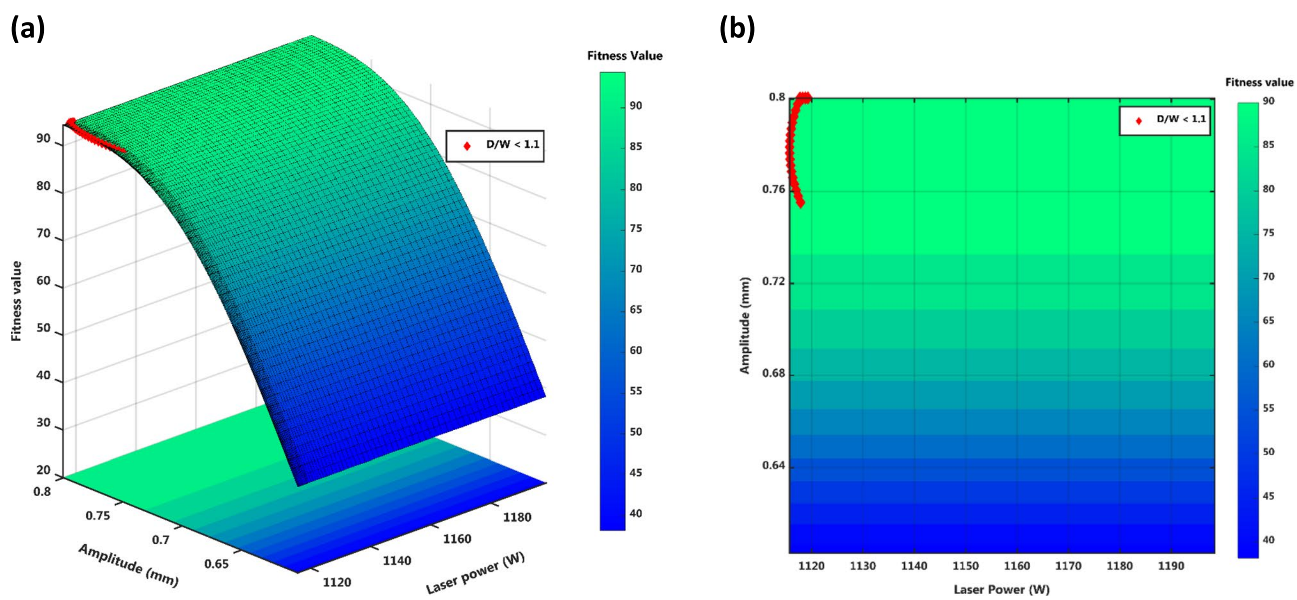


Fig. 14 **a** Surface of fitness function values for no-gap welding set-up. **b** Contour of fitness function values for no-gap welding set-up

7 Conclusion

This study investigated the effect of laser power, the wobbling amplitude, and the gap between plates during the laser welding of dissimilar aluminum alloys, namely, AA5052-H32 and AA6061-T6, using wobbling technique at low range of laser power compensated by a small spot diameter increasing the power density. The results of the weld characterization and optimization allow to draw the following conclusions:

- Weld imperfection investigation exhibits different levels of defect appearance relatively to the welding conditions. Underfill is the most recurrent imperfection in comparison to either undercut which is only visible for few experimental units or the root reinforcement ranged in the best quality for all weldments.
- The amplitude of oscillation has a huge influence on the weld shape, and the results of tensile test showed that breakage took place in the weld seam for all welding conditions. The ANOVA highlighted the significant effect of the gap on the weld tensile strength and suggested that no-gap welding is prompted to result in a strong weld.
- A correlation between the depth to width ratio and the ductility has been found, suggesting that the D/W should be kept as low as possible to enhance the elongation break.
- Response surfaces from regression model highlighted contradictory effects of parameters on ultimate tensile strength and the ductility justifying the necessity of parameters optimization.
- Parameter optimization results suggest setting the laser power at 1118 W and the amplitude at 0.8 mm with a no-gap configuration. This setting conducts to a depth to width ratio of 1.088, and the final mechanical properties are defined by ultimate tensile strength of 206.8 MPa and 4.5% of ductility. Those results are encouraging, and other mechanical properties such as fatigue life can be investigated through the same methodology.

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Author contribution Herinandrianina Ramiarison and Noureddine Barka are at the origin of this research project by proposing the approach and the methodology to address the problematic. Chris Pilcher with his team carried out all the welding experiments. Fatemeh Mirakhorli and François Nadeau brought their expertise in the aluminum field and metallurgy in results analysis. All authors discussed the results and contributed to the final manuscript.

Availability of data and material Data available on request from the authors.

Declarations

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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